



Nursing Practice Guideline

Arterial Line Management

Reference #: NPG34

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to Arterial Line Management. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

Table of Contents

Introduction	1
Principles.....	1
1. Insertion of Arterial Line	2
2. Maintenance of Arterial Line	4
3. Blood sampling	5
4. Line and Dressing Change	6
5. Removal of Arterial Line.....	7
Related Documents.....	8
References.....	8
Authorisation & Version Control	9

Principles

Observe the five moments for hand hygiene. Follow aseptic technique at all times and use the risk assessment outlined in the SCGOPHCG Infection Prevention and Management Aseptic Technique Policy. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.⁷

Patient identification must be checked when matching patient's identity to care, therapy and services.

All personnel who insert and/or manage arterial lines must adhere to the Work Health and Safety Act 2020

1. Insertion of Arterial Line

1.1 Medical Requirements

Experienced or supervised Medical Officer must insert the arterial cannula after obtaining verbal patient consent, if appropriate. Insertion should be documented in the inpatient notes and a Peripheral Vascular Catheter (PVC) Insertion and Observation Record must be commenced. Medical staff must prescribe the intraflow fluid.

1.2 Safety Information

Assessment of radial artery patency and collateral competency should be undertaken prior to arterial line insertion. Doppler ultrasonography (ensure sterile probe cover is used) is gold standard for this assessment ^{(1) Level 1}, alternatively an Allen's Test ^{(2) Level 1} could be performed by medical officer prior to radial line insertion ^{(3) Level 5}

Patients with arterial lines are not suitable to be managed outside of the following designated area:

- Intensive Care Unit,
- Emergency Department,
- Post Anaesthetic Care Unit (PACU)
- Respiratory (ReSCU) Special Care Unit, Neurosurgical HDU, Coronary Care Unit, Cardiovascular Interventional Lab (CVIL) Procedural Room and Endovascular Neurological Unit

If found outside of these areas, please contact CNS/AHCNS for advice

1.3 Nursing Management

During catheter insertion, assistance may be required to restrain and position the insertion site. A rolled towel and tape may be used for this purpose. Prior to insertion of cannula:

- a) Prime arterial pressure monitoring line; with 500ml bag of Sodium Chloride ^{(4) Level 1}
- b) Place IV bag into a pressure bag; inflate pressure bag to 300mmHg ^{(2) Level 5}
- c) Ensure the line used has an access tap insitu, change tap port to non-vented cap once primed.

Document the site and date of insertion on the ICU Flow Chart, Emergency Department Nursing Admission Record and/or Peripheral Vascular Catheter (PVC) Insertion and Observation Record.

1.4 Procedural Information

Requirements

- Pressure Bag
- Arterial Pressure Monitoring line
- 500ml Normal Saline 0.9%
- Pressure cable
- Cannula of choice
- Sterile gel and probe cover
- Sterile Dressing Pack
- Chlorhexidine 2% with Alcohol 70% solution swab sticks ^{(1) Level 3}
- 1 x 26G needle

- 1 x Drawing up needle
- 1 x 5ml syringe
- Lignocaine 1%
- Red non vented cap (packs come with white sterile non-vented caps that may also be used)
- Chlorhexidine impregnated dressing (e.g., 3M Tegaderm CHG) (5) Level 5.

For securement of radial line:

- Arm Board
- Combine x 3
- Leukoplast (or similar) tape
- Labels (Arterial) (1) Grade B
- PPE including plastic apron, eye protection and non-sterile gloves
- Sterile Gloves
- Wet strength bag

Procedure

Post Insertion

1. Perform hand hygiene.
2. Place chlorhexidine impregnated dressing over inserted cannula (do not cover any part of the line) using a non-touch technique and then use tape to support line so as to prevent drag (1) Grade B. For radial lines, loop line around thumb to reduce risk of line kinking. See Figure 1.
3. Use arm board (padded with combine) to ensure optimal cannula position. Ensure that the line is not causing pressure to skin between index finger and thumb. See Figure 1.
4. To secure the arm board use Leukoplast tape padded with combines.
5. Perform hand hygiene.



Figure 1: Optimal positioning using arm board

2. Maintenance of Arterial Line

2.1 Safety Information

Aseptic technique is to be maintained at all times during maintenance of the arterial line.

- Cannula insertion site should be visible at all times.
- Ensure all connections are secure and all caps are non-vented (red bung).
- Line must be transduced continuously with waveform visible on monitor at all times.
- Keep insertion site immobilised with dressings and a padded splint (which must be removed daily or PRN to check for pressure areas).
- Set Systolic blood pressure and Mean blood pressure high and low alarms within 20% of patient's actual blood pressure.
- Alarms must never be disabled.
- Ensure the line is free of air. ^{(6) Level 1}
- Nothing is to be infused or administered via Arterial Line (aside from Intraflow).
- Two Nurses are required for any line dressing / changes.

2.2 Nursing Management

- Lines must be labelled with 'Intra-Arterial' sticker ^{(1) Level 5} Lines (including fluid) and dressings are to be changed every 72 hours, or with a change in cannula.
- Using aseptic non touch technique, zero line to atmospheric pressure ^{(2) Level 5} after insertion, at beginning of shift ^{(2) Level 5} and prn ^{(7) Level 5} Turn three-way tap "off" to patient, remove non-vented cap so tap port is open to air; zero arterial line on monitor, wait for the "0" to appear, then turn tap back to the off position and replace cap. Check arterial line trace on monitor for accuracy.
- Perform a Square Wave test once per shift ^{(6) Level 5} The optimal time for this is after the line has been zeroed. To do this: flush the system via the intraflow for one second, observe for square waveform on monitor, followed by a brief undershoot and oscillation back to the baseline. The monitor should display the same pressure as the pressure bag ^{(2) Level 5}.
- Select an appropriate monitor scale range (inclusive of zero point) for arterial pressure to ensure reliable values and a visual waveform at all times. Avoid the use of 'optimum scale' setting which is available on some monitors.
- Check manual / NIBP blood pressure (BP) on the same arm at commencement of shift and as necessary to check accuracy of arterial line. Refer to medical staff if difference >10%. Differentiate readings by recording manual BP reading in red pen and arterial line reading in blue or black on ICU Flow Chart (MR870) or ORC chart. Write 'Manual BP' against the red reading. Remove NIBP / manual cuff once check complete.
- Perfusion distal to line insertion site must be assessed 4/24 by checking capillary refill and colour and warmth of extremities distal to the arterial cannula ^{(7) Level 5}.
- Document findings on ICU Flow Chart (MR870) or Neurovascular Observation Chart (MR 873) and in nursing notes with patient assessment.
- Ensure there is sufficient fluid in the attached 500 ml intravenous (IV) bag of 0.9% Sodium Chloride ^{(4) Level 5}, that the pressure bag is inflated to 300mmHg (to deliver

- 3ml/hr) ^{(2) Level 5}, and that all three-way taps are positioned to allow free flow of fluid through the system.
- Level air port of transducer at phlebostatic Axis ^{(2, 3, 7) Level 5} (junction of 4th intercostal space and mid-axilla line) if transducer is mounted on transducer holder.
- Maintain transducer as close as possible to catheter tip if not mounted (see figure 1). ^{(6) Level 5}
- Monitor and record Temperature 4/24 and document on ORC Chart or ICU Flow Chart (MR 870).
- Observe for signs of infection around cannula insertion site ^{(3, 7) Level 5}. Document PIVAS score on Peripheral Vascular Catheter (PVC) Insertion and Observation Record chart (MR864.2), or ICU Flow Chart (MR870) once per shift.
- If air is seen in line, using ANTT principles, attach syringe to three-way tap at transducer then aspirate fluid until the air bubble has been collected in the syringe. Using the intraflow fluid; flush the access port of the three-way tap. Remove the syringe and discard. Replace the cap and flush the line using intraflow fluid to ensure all blood has been returned to the patient.

3. Blood sampling

3.1 Procedural Information

Requirements

- Large 70% alcohol wipe
- Sterile gauze
- 2ml syringe
- Blood Tubes as required
- Blood sampling device (vacutainer)
- PPE including safety glasses/visor and non-sterile gloves
- Red non-vented cap

Procedure

1. Confirm patient identity and explain the procedure. ^{(8) Level 5}
2. Suspend monitor alarms.
3. Perform hand hygiene and apply PPE.
4. Using non touch technique, scrub area around non vented cap; with a large 70% alcohol- wipe using friction and allow to air dry for 30 seconds.
5. Remove non-vented cap on 3-way tap of Arterial Line and discard. Attach 2ml syringe, open 3 way tap to patient, and withdraw 3mls of dead space volume (*flush solution + diluted blood = potential contamination of results*) from sample line.
6. Turn 3-way tap off to patient, remove and discard syringe.
7. Replace with blood sampling device.
8. Open to patient and fill required blood tubes, in correct order (as per PathWest information).
9. Turn 3-way tap off to patient, remove blood sampling device. Gently roll and invert blood tubes to mix contents.
10. With 3-way tap off to patient, flush sample port using intraflow fluid into sterile gauze to remove blood.
11. Replace Red non-vented cap on 3-way tap.

12. With 3-way tap open to patient, flush sample line using quick short flushes of intraflow fluid until sample line is clear (minimises discomfort to patient and risk of arterial spasm at cannula site).
13. Check arterial waveform and activate monitor alarms.
14. Label filled blood tubes before leaving bed space.
15. Arterial blood gas samples should be analysed within 15 minutes if held at room temperature ^{(9) Level 5}.
16. Dispose of rubbish and sharps.
17. Remove PPE and perform hand hygiene.

4. Line and Dressing Change

4.1 Procedural Information

Requirements

- Sterile Dressing Pack
- PPE including plastic apron, safety glasses or visor:
- Sterile Gloves (for nurse changing line)
- Non-sterile gloves (For nurse assisting)
- Chlorhexidine impregnated dressing (e.g., 3M Tegaderm CHG)
- Tape (Leukoplast)
- Arm splint (for radial lines)
- 500ml bag 0.9% Normal Saline
- Arterial Pressure Monitoring Kit
- Pressure Bag
- Chlorhexidine 2% in Alcohol 70% solution swab sticks
- Large 70% alcohol-wipe
- Intra-arterial line label
- Wet strength bag

Procedure

NB: Two Nurses are required for this procedure

1. Perform hand hygiene
2. Clean trolley using combined detergent and disinfectant wipe. Allow to dry.
3. Gather equipment.
4. Perform hand hygiene
5. Prime new arterial pressure monitoring kit, ensuring all ports are free of air.
6. Open dressing pack using corners and empty Chlorhexidine 2% with Alcohol 70% solution swab stick onto sterile area.
7. Perform hand hygiene.
8. Don non-sterile gloves and PPE.
9. 'Nurse one' suspends monitors alarms and then supports patients' hand while second nurse removes existing dressing while supporting cannula.
10. 'Nurse two' performs hand hygiene and don sterile gloves.
11. Cleanse skin liberally around insertion site with Chlorhexidine solution swab stick for 30 seconds. Allow to air dry for 30 seconds.
12. Scrub the hub with a large 70% alcohol wipe using friction and allow to air dry for 30 seconds.

13. Nurse one occludes radial artery proximal to insertion site ensuring waveform is lost. Nurse two secures cannula in one hand and disconnects pressure line, immediately replacing with new line.
14. Nurse one releases occlusive pressure and connects transducer cable to new line.
15. When adequate waveform is established, apply chlorhexidine impregnated dressing ^{(5) Level 5} over insertion site using a non-touch technique and secure (Figure 1).
16. Dispose of all sharps and waste appropriately.
17. Remove PPE
18. Perform hand hygiene.
19. Complete documentation on ICU flow chart and relevant Nursing Care Plan.

5 Removal of Arterial Line

5.1 Medical Requirements

Verbal medical order should be obtained prior to removal of cannula. Arterial cannula to be changed by medical staff only if cannula suspected to be cause of sepsis if site is inflamed or localised infection is suspected.

5.2 Safety Information

Aseptic technique is to be maintained at all times during removal of the arterial line.

5.3 Nursing Management

- Document the removal of cannula in the inpatient notes, and on the relevant Nursing Care Plan. Peripheral Vascular Catheter (PVC) Insertion and Observation Record (MR 864.2)
- Femoral Arterial Sheath removal post cardiac catheter procedure must follow the relevant local area Standard Operating Procedure.

5.4 Procedural Information

Requirements

- Sterile Dressing Pack
- Chlorhexidine 2% with Alcohol 70% solution swab sticks ^{(1) Level 3}
- Sterile Gauze
- Elastoplast to create pressure dressing
- PPE (Non-sterile gloves, plastic apron, safety glasses / visor)

Procedure

1. Explain procedure to patient ^{(10) Level 5}.
2. Clean trolley using combined detergent and disinfectant wipe. Allow to dry.
3. Gather equipment. Perform hand hygiene and apply PPE.
4. Open dressing pack using corners and empty Chlorhexidine 2% with Alcohol 70% solution swab stick onto sterile area. Prepare pressure dressing (Usually Elastoplast with sterile gauze) ^{(3) Level 5}.
5. Perform hand hygiene.
6. Turn stopcock off to patient.
7. Disconnect Transducer.
8. Perform hand hygiene and don non-sterile gloves.
9. Remove occlusive chlorhexidine impregnated dressing.
10. Remove gloves.
11. Perform hand hygiene and don non-sterile gloves.

12. Using non-touch technique clean around insertion site using chlorhexidine solution swab stick. Allow to air dry for 30 seconds
13. Apply pressure with sterile gauze just proximal to catheter insertion site.
14. Remove cannula and inspect to ensure intact ⁽³⁾ Level 5.
15. Apply digital pressure for a minimum of 5 minutes or until bleeding has ceased ⁽³⁾ Level 5.
16. Apply pressure dressing and leave insitu for 24 hours.
17. Dispose of waste.
18. Remove PPE and perform hand hygiene.
19. Document removal of line on the Peripheral Vascular Catheter (PVC) Insertion and Observation Record (MR864.2).
20. Continue to observe site, for bleeding and for distal circulation.

Related Documents

Legislation:

- Health Practitioner Regulation National Law (WA) Act 2010
- Work Health and Safety Act 2020
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses March 2018 Policy:
- NMHS Clinical Documentation Policy
- SCGH Patient Identification and Procedure Matching Policy
- SCGOPHCG Infection Prevention and Management Manual - Aseptic Technique Policy
- WA Department of Health Mandatory Policy 0131/20 High Risk Medication Policy .
- Australian Commission on Safety and Quality in Health Care. National Standard for User applied Labelling of Injectable Medicines, Fluids and Lines. Sydney: ACSQHC, 2015.

Guidelines:

- SCGOPHCG Clinical Practice Guideline Peripheral Intravenous Catheters (PVC) -Insertion, Maintenance and Removal.

References

1. Madhavi, Sai Sivapuram. Arterial Lines: Insertion. JBI Evidence Summary. Ovid; 2020. Accessed April 04, 2023. https://ovidsp-dc1-ovid-com.gelibresources.health.wa.gov.au/ovid-b/ovidweb.cgi?&S=FJHDFPPPAPACKMIPKPLJHHEMAADJAA00&Link+Set=S.sh.21%7c2%7csl_190
2. Sandeep, Moola. Arterial Lines: Monitoring and Transducers. JBI Evidence Summary. Ovid; 2021. Accessed April 04, 2023. https://ovidsp-dc1-ovid-com.gelibresources.health.wa.gov.au/ovid-b/ovidweb.cgi?&S=KNEBFPNPIKACKMOKKPLJKGPMMLLPAA00&Link+Set=S.sh.21%7c2%7csl_190
3. Theodore AC, Clermont G, Dalton A. Intra-arterial catheterization for invasive monitoring: Indications, insertion techniques, and interpretation. UpToDate. Wolters Kluwer. Updated November 02, 2022. Accessed April 04, 2023. <https://www.uptodate.com/contents/intra-arterial-catheterization-for-invasive-monitoring-indications-insertion-techniques-and-interpretation?csi=e10946da-fe38-462e-ab46-444ebc75b647&source=contentShare>
4. Moola, Sandeep. Arterial Lines: Flushing Media. JBI Evidence Summary. Ovid; 2021. Accessed April 04, 2023. https://ovidsp-dc2-ovid-com.gelibresources.health.wa.gov.au/ovid-b/ovidweb.cgi?&S=JLHAFPMHENEKENKFIPLJBHPFONLFAA00&Link+Set=S.sh.21%7c7%7csl_190
5. Porritt K. Peripheral Arterial Catheters: Dressings and catheter securement. JBI Evidence Summary. Ovid; 2021. Accessed April 04, 2023. https://ovidsp-dc2-ovid-com.gelibresources.health.wa.gov.au/ovid-b/ovidweb.cgi?&S=JLHAFPMHENEKENKFIPLJBHPFONLFAA00&Link+Set=S.sh.21%7c1%7csl_190
6. Arterial Line: Insertion. JBI Recommended Practice. Ovid; 2020. Accessed April 04, 2023. https://ovidsp-dc2-ovid-com.gelibresources.health.wa.gov.au/ovid-b/ovidweb.cgi?&S=FNBBFPMGCLEBENCEJPLJFEIHBFAIAA00&Link+Set=S.sh.21%7c7%7csl_190
7. Arterial Line: Management. JBI Recommended Practice. Ovid; 2021. Accessed April 04, 2023. https://ovidsp-dc2-ovid-com.gelibresources.health.wa.gov.au/ovid-b/ovidweb.cgi?&S=FNBBFPMGCLEBENCEJPLJFEIHBFAIAA00&Link+Set=S.sh.21%7c3%7csl_190

8. Arterial Line: Blood sampling using a needleless device or syringe. JBI Recommended Practice. Ovid; 2021. Accessed April 04, 2023. https://ovidsp-dc2-ovid-com.qelibresources.health.wa.gov.au/ovid-b/ovidweb.cgi?&S=FNBBFPMGCLEBENCEJPLJEFIHBAFIAA00&Link+Set=S.sh.21%7c5%7csl_190
9. Theodore AC. Arterial Blood Gases. UpToDate. Wolters Kluwer. Updated August 10, 2022. Accessed April 04, 2023. https://www-uptodate-com.qelibresources.health.wa.gov.au/contents/arterial-blood-gases?search=arterial%20blood%20gases&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
10. Arterial Line: Removal. JBI Recommended Practice. Ovid; 2020. Accessed April 04, 2023. https://ovidsp-dc2-ovid-com.qelibresources.health.wa.gov.au/ovid-b/ovidweb.cgi?&S=FNBBFPMGCLEBENCEJPLJEFIHBAFIAA00&Link+Set=S.sh.21%7c2%7csl_190
11. Madhava SS. Arterial Lines: Insertion. JBI Evidence Summary. Ovid; 2020. Accessed April 04, 2023. https://ovidsp-dc2-ovid-com.qelibresources.health.wa.gov.au/ovid-b/ovidweb.cgi?&S=MPDIFPAGHAEBENANIPLJPEPFECNIAA00&Link+Set=S.sh.21%7c3%7csl_190

JBI Levels of Evidence

Level 1 Experimental designs

Level 3 Observational designs

Level 5 Expert opinion and bench research

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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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